DR-EX026: SQLDB failure on AZ1TMBDBVWPR201 (INST1)

Overview of SQL Server Architecture

`AZ1TMBDBVWPR201.corp.teranet.ca` and `AZ1TMBDBVWPR202.corp.teranet.ca` are nodes in a two-node SQL Server Availability Group named `az1tmbdbvwprag.corp.teranet.ca`, in a High-Availability configuration with synchronous commit between each node. Either node can operate as the principal node serving databases with the Availability Group configured for automatic failover.

`AZ3TMBDBVWPR203.corp.teranet.ca` is in a single-node Availability Group named `az3tmbdbvwprag.corp.teranet.ca`. Data in this Availability Group is typically less than 100ms, and no more than a few seconds “behind” data in the `az1tmbdbvwprag.corp.teranet.ca` Availability Group during normal operation.

There is a SQL Server Distributed Availability Group with asynchronous commit between the `az1tmbdbvwprag` and `az3tmbdbvwprag` Availability Groups, configured for manual failover between the two groups for Disaster Recovery purposes.

If `AZ1TMBDBVWPR201` goes offline, `AZ1TMBDBVWPR202` will automatically take over as the principal node serving databases to TMB applications seamlessly (and the opposite is also true allowing for automated fail back). Applications are configured to connect to SQL Server via a DNS CNAME named `mb-db.corp.teranet.ca` which points at `az1tmbdbvwprag.corp.teranet.ca` during normal, non-disaster recovery, operation. During a DR exercise or a real DR event, that DNS CNAME should be repointed to `az3tmbdbvwprag.corp.teranet.ca`.

Planned Disaster Recovery Exercise

During a DR Exercise, general steps are:

1. Switch Availability Mode from asynchronous commit to synchronous commit.
2. Ensure all data is synchronized between the principal AG (az1tmbdbvwprag), and the forwarder AG (az3tmbdbvwprag) for all databases that are part of the AG. Confirm LSNs match for all AG databases on both the Distributed AG Primary and Forwarder.
3. Disable client access to the Distributed AG.
4. Failover from the primary AG to the forwarder AG.
5. Request a DNS change for the `mb-db.corp.teranet.ca` CNAME to reference the new Distributed AG Primary name (i.e. `az3tmbdbvwprag.corp.teranet.ca` instead of `az1tmbdbvwprag.corp.teranet.ca`)
6. Switch the Distributed AG commit mode to asynchronous.

The above general steps are encapsulated into a set of T-SQL scripts, which are attached to this page.

DAG_Failover_mb-db-in-azure.zip

Inside the `DAG_Failover_mb-db-in-azure.zip` file are several Windows `.cmd` scripts:

1. `check-sync.cmd` which can be used to check the synchronization status of the two Availability Groups, `az1tmbdbvwprag.corp.teranet.ca` and `az3tmbdbvwprag.corp.teranet.ca`
2. `failover_from_primary_to_secondary.cmd` for failing over the Distributed AG from `az1tmbdbvwprag.corp.teranet.ca` to `az3tmbdbvwprag.corp.teranet.ca`
3. `failover_from_secondary_to_primary.cmd` for failing over the Distributed AG from `az3tmbdbvwprag.corp.teranet.ca` to `az1tmbdbvwprag.corp.teranet.ca`
4. The various `.sql` scripts in the folder are called as-needed by the `.cmd` scripts, and should not be modified without a clear understanding of the purpose.

Unplanned Response to a Disaster

- If SQL Servers in the primary Availability Group `az1tmbdbvwprag.corp.teranet.ca` are still available, steps are the same as during a DR Exercise above.

- If the primary SQL Server Availability Group `az1tmbdbvwprag.corp.teranet.ca` is unavailable for any reason, and there is a need to bring up the Disaster Recovery Availability Group `az3tmbdbvwprag.corp.teranet.ca` the following steps should be taken:

1. Connect to the node in the DR AG. If connecting via sqlcmd:

```
sqlcmd.exe -S az3tmbdbvwpr203.corp.teranet.ca\TMB -M -d master -E -b
```

2. Failover the Distributed Availability Group using the `FORCE_FAILOVER_ALLOW_DATA_LOSS` option.

```sql
ALTER AVAILABILITY GROUP [mb-db-in-azure]
FORCE_FAILOVER_ALLOW_DATA_LOSS;
```

3. Use the following query in SSMS to check the status of the Availability Group(s) after failover completes:

```sql
SELECT
      [ag].[name]
    , [ag].[is_distributed]
    , [ar].[availability_mode_desc]
    , [ar].[replica_server_name]
    , [ar].[failover_mode_desc]
    , [ars].[synchronization_health_desc]
    , [ars].[connected_state_desc]
    , [ars].[operational_state_desc]
    , [ars].[recovery_health_desc]
    , [ars].[role_desc]
    , [ars].[last_connect_error_description]
FROM sys.availability_groups ag
    INNER JOIN sys.availability_replicas ar ON ag.group_id = ar.group_id
    INNER JOIN sys.dm_hadr_availability_replica_states ars
       ON ag.group_id = ars.group_id
            AND ar.replica_id = ars.replica_id;
```

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